

Computer Laboratory Usage Monitoring System

ASP.NET Core MVC Laboratory Exercise Specification

Objective

Develop an ASP.NET Core MVC application for monitoring student computer laboratory usage using Model Binding, Cookie Authentication, Data Annotation Validation, Repository Pattern (in-memory), Razor Views, and Bootstrap.

Business Scenario

A school computer laboratory replaces its paper logbook with a web application. Laboratory assistants register students, assign computers, monitor active sessions, and record time out.

Learning Objectives

MVC, Model Binding, DTOs, Validation, Cookie Authentication, Authorization, Repository Pattern, CRUD, Search, Bootstrap.

Functional Requirements

Register User, Login, Logout, Register Laboratory Session, View Monitoring List, Edit Session, View Details, Record Time Out, Search.

Session Fields

Session Number (auto-generated), Student Number, First Name, Last Name, Course, Year Level, Computer Number, Purpose, Time In, Time Out, Status, Notes.

Monitoring

Display Session Number, Student, Computer, Time In, Time Out, Status, Actions.

Status

Using, Finished.

Validation

Use Required, Range, Display, DataType and other Data Annotation attributes. Validate ModelState.

Authentication

Public: Login/Register. Protected: Laboratory pages with Cookie Authentication and [Authorize].

Entities

User: Id, FirstName, LastName, Email, Username, Password. LaboratorySession: Id, SessionNumber, StudentNumber, FirstName, LastName, Course, YearLevel, ComputerNumber, Purpose, TimeIn, TimeOut, Status, Notes.

Repositories

UserRepository and LaboratorySessionRepository using static List.

Controllers & Views

Controllers: HomeController, AccountController, LaboratorySessionController. Views: Login, Register, Index, Create, Edit, Details, TimeOut.

Submission

Complete MVC application with authentication, CRUD, validation, search, Bootstrap UI, repository pattern, and clean code.